Practical 9: Commercial Bank Real Estate Loans Forecast

Objective

The goal of this analysis is to forecast the monthly volume of commercial bank real estate loans for the next 20 months using the `bank_case.txt` dataset.

Dataset

- **File**: `bank_case.txt`
- **Observations**: 70 monthly records
- **Variable**: Volume of commercial loans (Billions)

Methodology

1. **Time Series Conversion**: The data was converted into a monthly time series object (`ts`).
2. **Exploratory Data Analysis**: Visualized the trend and seasonality.
3. **Stationarity Testing**: Performed KPSS test and analyzed ACF/PACF plots.
4. **Model Selection**: Used `auto.arima()` to identify the optimal (S)ARIMA model based on AIC/BIC.
5. **Forecasting**: Generated a 20-month forecast with 80% and 95% confidence intervals.
6. **Diagnostics**: Checked residuals for white noise properties using the Ljung-Box test.

Code Snippet

The analysis followed a structured approach, utilizing the `forecast` and `tseries` libraries:

1. Data Preparation & Time Series Object

```
# Loading the commercial loan volume from a text file
bank_data <- read.table("../Practical7/bank_case.txt", header = FALSE)
# Converting to a monthly time series (frequency=12)
bank_ts <- ts(bank_data$V1, frequency = 12, start = c(1, 1))
```

2. Exploratory Data Analysis & Decomposition

```
# Plotting the original series to identify components
plot(bank_ts, main = "Commercial Bank Real Estate Loans")
# Additive decomposition to observe the strong trend
bank_decomp <- decompose(bank_ts, type = "additive")
plot(bank_decomp)
```

3. Stationarity Testing

```
# KPSS test for level stationarity
kpss.test(bank_ts, null = "Level")
# ACF/PACF analysis
acf(bank_ts, lag.max = 48)
pacf(bank_ts, lag.max = 48)
```

4. Optimal Model Selection (auto.arima)

```
# Use auto.arima to find the best (S)ARIMA parameters based on minimized
AIC/BIC
best_model <- auto.arima(bank_ts, seasonal = TRUE, stepwise = FALSE,
approximation = FALSE)
summary(best_model)
```

5. Residual Diagnostics & Goodness of Fit

Before forecasting, we must ensure the model is statistically sound by checking if the residuals behave like white noise:

```
# Checkresiduals function performs Ljung-Box test and plots residual
distribution
checkresiduals(best_model)
```

6. Forecasting as a Function

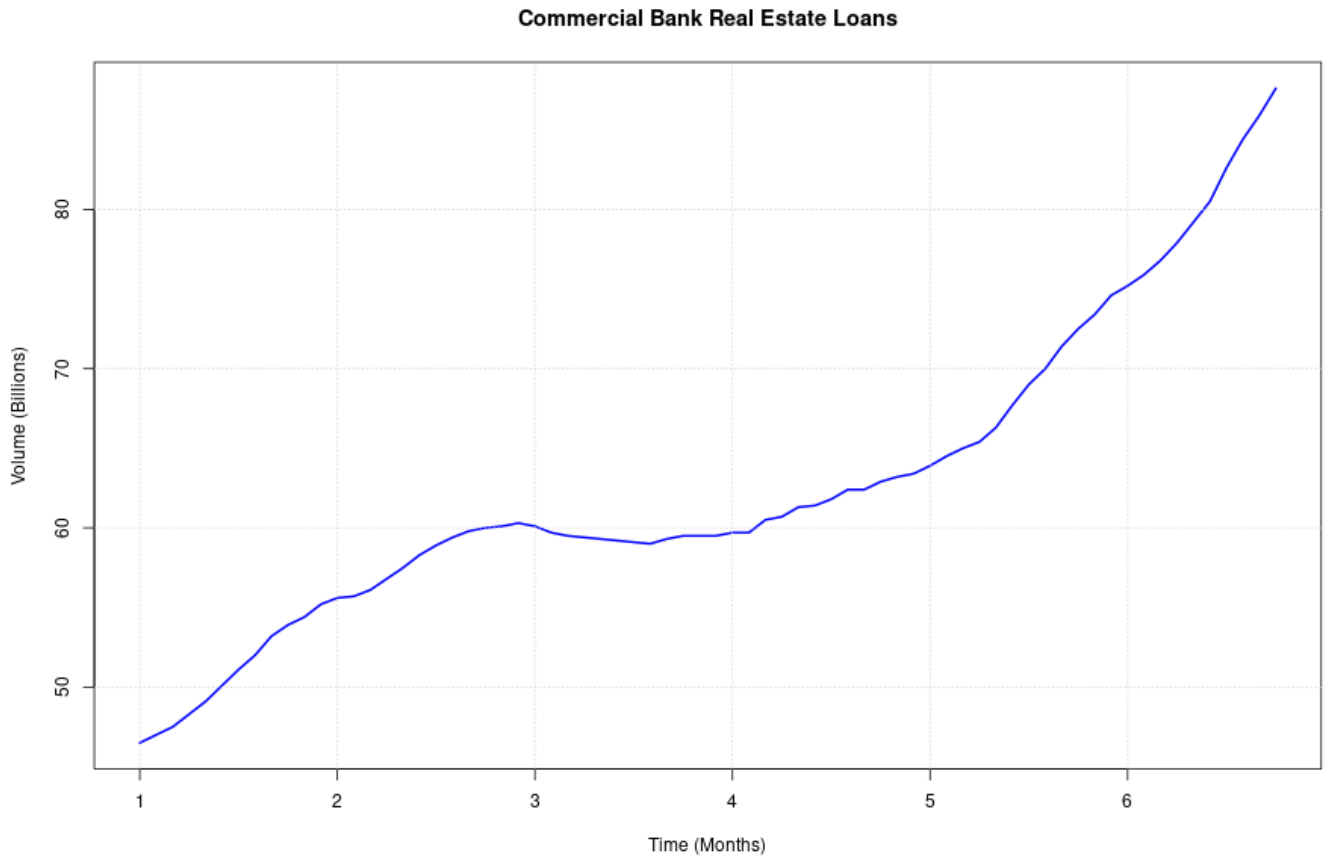
Once the model's goodness of fit is confirmed, the primary forecasting function is `forecast()`, which takes the fitted model object and the horizon `h` (number of periods) as arguments:

```
# Forecasting the next 20 months (h=20)
bank_forecast <- forecast(best_model, h = 20)
# Visualizing the forecast including confidence intervals
plot(bank_forecast)
```

Analysis and Results

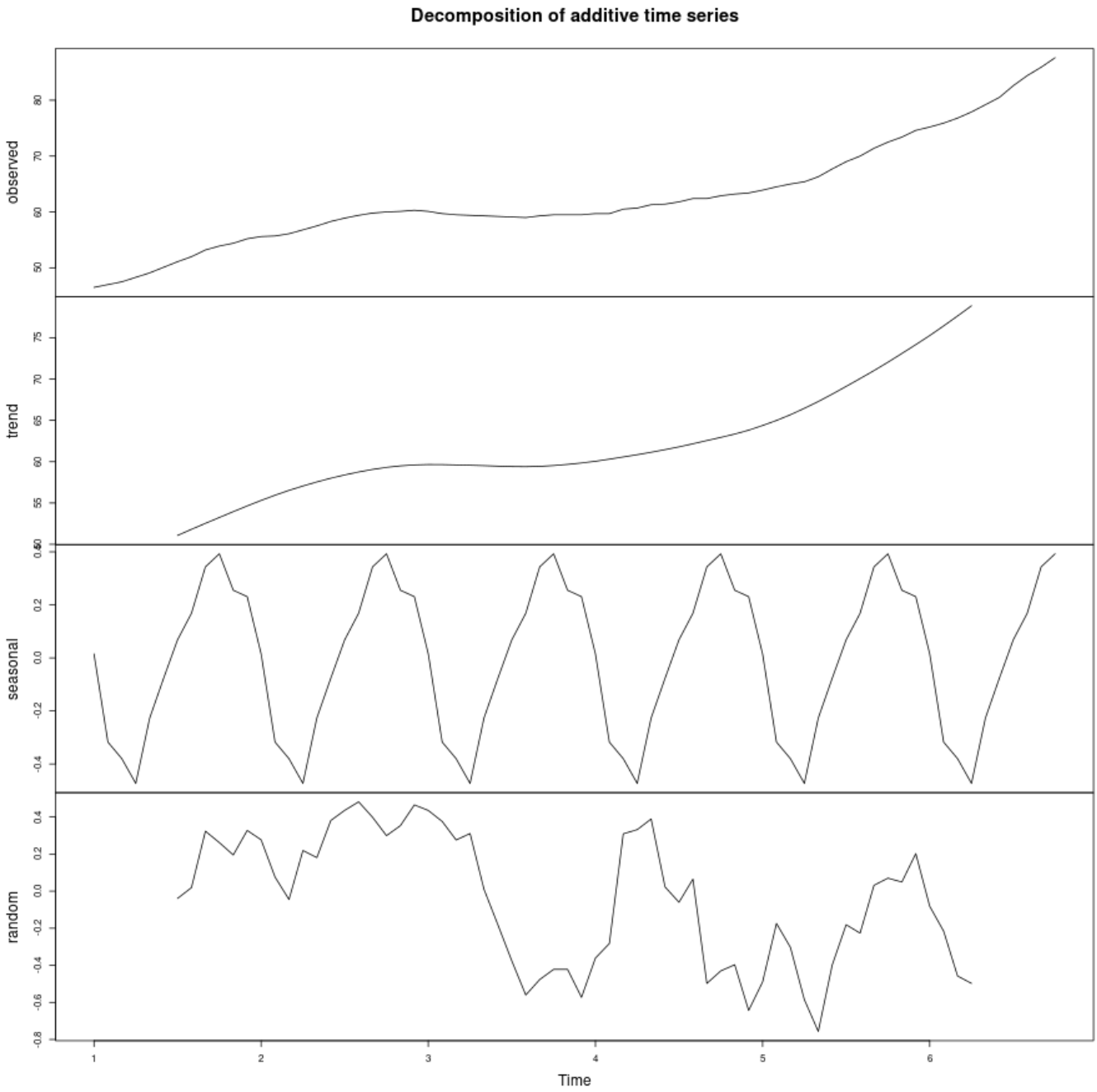
1. Data Visualization

The original series shows a clear non-linear upward trend, especially in the latter half of the dataset.



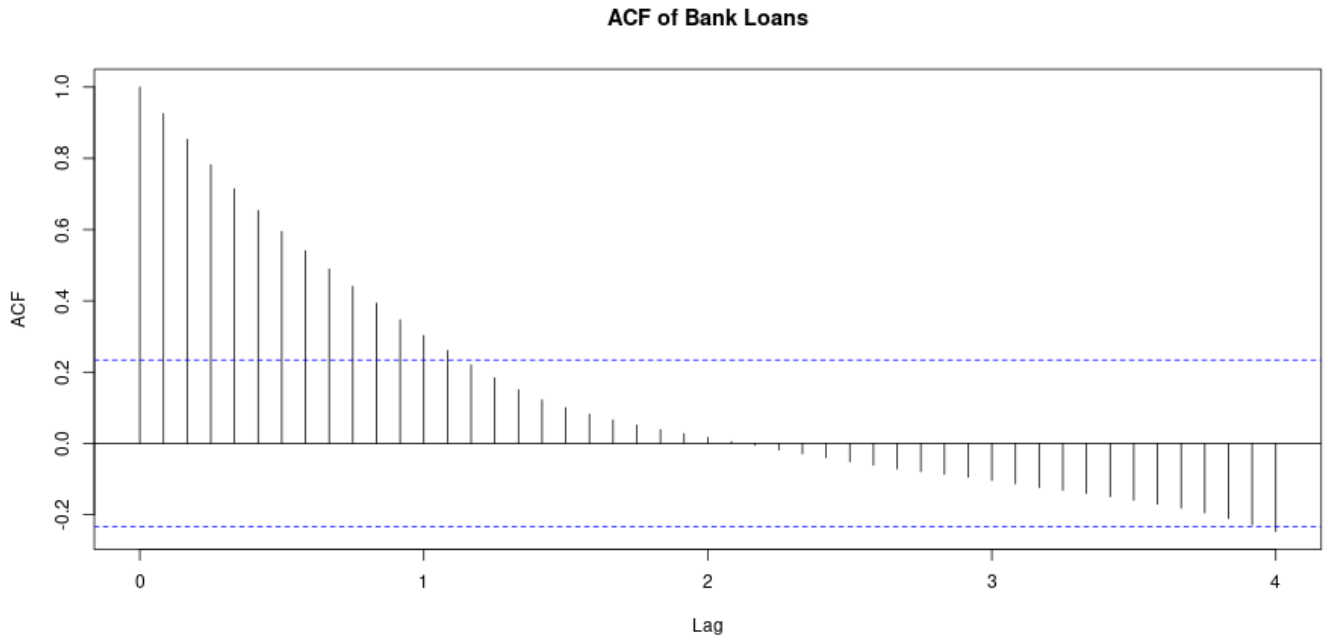
2. Decomposition

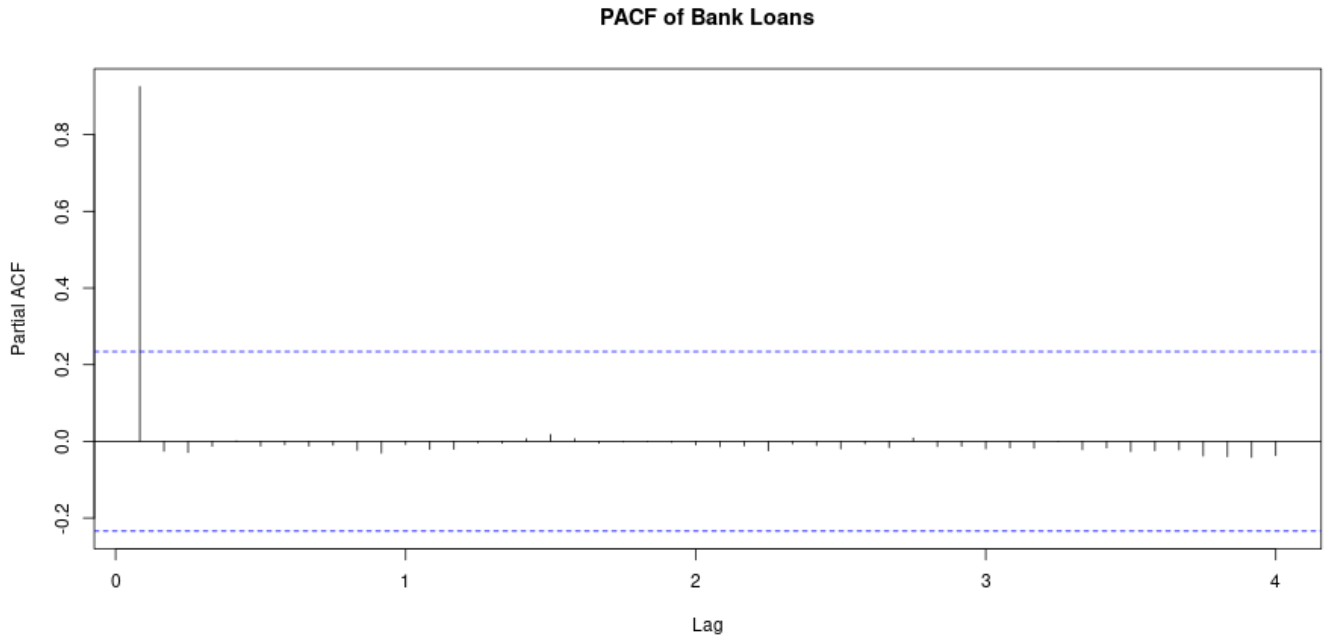
The decomposition shows a strong trend component.



3. Stationarity and ACF/PACF

- **KPSS Test**: Level stationarity was rejected ($p < 0.01$), indicating the series is non-stationary.
- **ACF**: Showed very slow decay, confirming a trend.





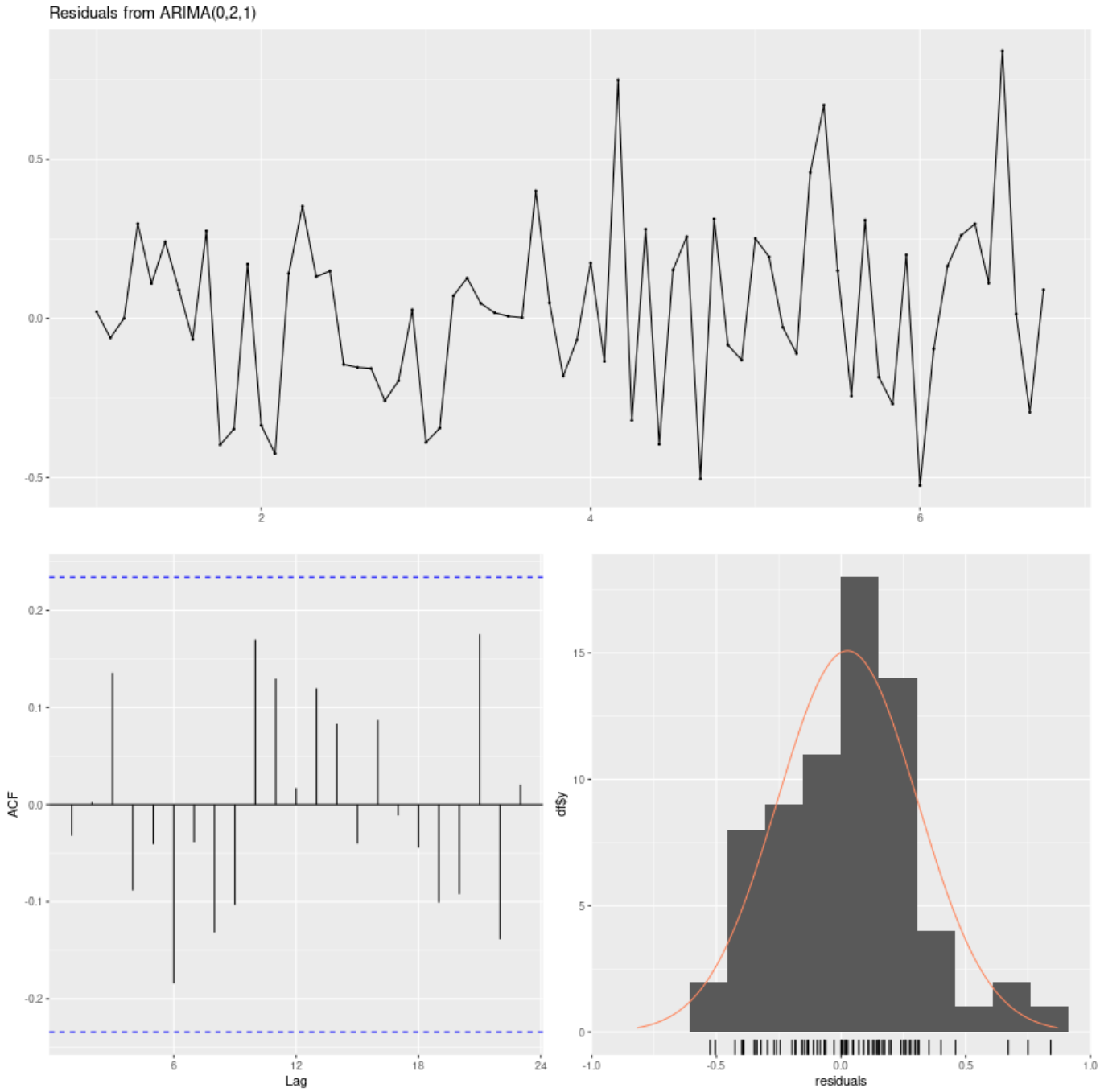
4. Model Selection

The `auto.arima` function selected an **ARIMA(0,2,1)** model.

- **Model**: Second-order differencing ($d=2$) with a first-order moving average ($MA(1)$).
- **AIC**: 26.13
- **BIC**: 30.57

5. Residual Diagnostics (Goodness of Fit)

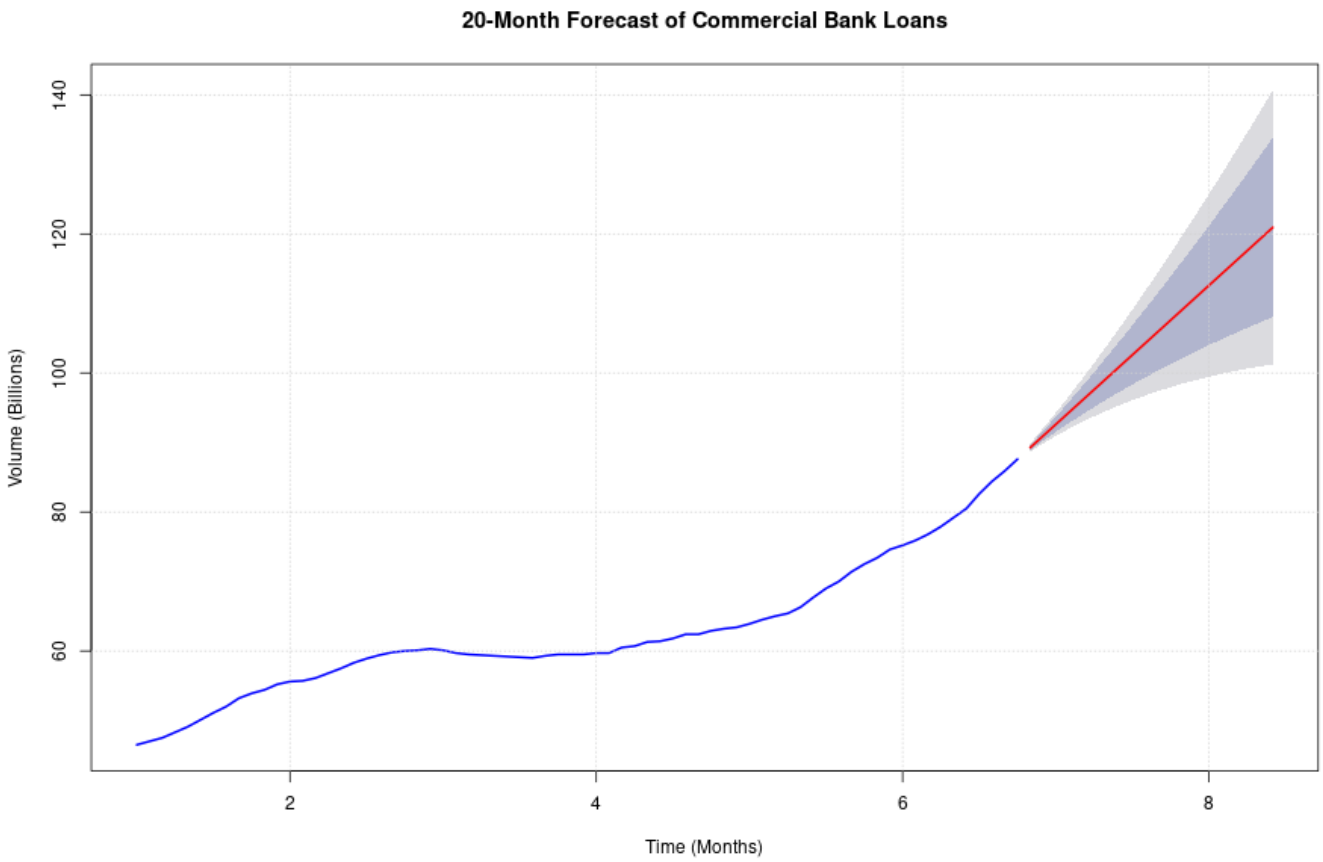
The Ljung-Box test resulted in a p-value of **0.4446**, which is greater than 0.05. This indicates that the residuals are indistinguishable from white noise, meaning the model has captured the underlying pattern well and is suitable for forecasting.



6. Forecast (Next 20 Months)

The forecast predicts a continued upward linear trend for the next 20 months.

Period	Month-Year	Point Forecast	95% Lower Bound	95% Upper Bound
+1 Month	Nov 6	**89.27**	88.70	89.83
+6 Months	Apr 7	**97.60**	93.77	101.43
+12 Months	Oct 7	**107.60**	97.96	117.23
+18 Months	Apr 8	**117.60**	100.62	134.58
+20 Months	Jun 8	**120.93**	101.21	140.65



Projections and Key Insights

- **Trend Persistent**: The ARIMA(0,2,1) model with $d=2$ implies that the series is non-stationary and the trend is very persistent, leading to a strong linear projection.
- **Widening Uncertainty**: The confidence intervals widen significantly over time (from ± 0.5 in month 1 to ± 20 in month 20), indicating that while the trend is upward, the long-term volume is subject to increased market volatility.
- **Model Validity**: The residual diagnostics confirm the model is statistically sound for this dataset.

Conclusion

The volume of commercial bank real estate loans is expected to continue its growth. The ARIMA(0,2,1) model provides a statistically sound fit for the data, as evidenced by the residual diagnostics.